

Notes: Gorton and Pennacchi (JF, 1990)

Financial Intermediaries and Liquidity Creation

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Model setup	3
3.1	Dates, agents, and preferences	3
3.2	Endowments and technology	3
3.3	Information structure	4
4	Full-information stock-market benchmark	4
5	Asymmetric information and insider trading	4
5.1	The insider coalition	4
5.2	Proposition 1: insider trading equilibrium	5
6	Private liquidity creation	5
6.1	Bank intermediation and safe debt	5
6.2	Breaking the insider coalition	5
6.3	Intermediary equilibrium	6
7	Corporate debt and nonbank liquidity creation	6
8	Government intervention	6
8.1	Deposit insurance	6
8.2	Government debt market	7
9	Theoretical identification and design	7
10	Extracted paper visuals and original equation panels	8
10.1	Extract 1: Original title and abstract - information-insensitive securities	8
10.2	Extract 2: Full-information pricing equations (1)-(5) - no-storage and storage prices	8
10.3	Extract 3: Proposition 1 - Insider Trading	9
10.4	Extract 4: Bank intermediation contract - riskless debt condition (14)-(16)	9

10.5 Extract 5: Proposition 2 - Nonexistence of stock-market insider equilibrium	10
10.6 Extract 6: Proposition 3 - Existence of an intermediary equilibrium	10
10.7 Extract 7: Corporate debt and government-debt introduction - nonbank liquidity creation	11
10.8 Extract 8: Proposition 4 - Deposit Insurance	11
10.9 Extract 9: Conclusion - banks, safe debt, Treasury securities, and money market funds	12
11 Short summary	13
12 Conclusion	14
12.1 Core conclusion	14
12.2 Economic intuition	14
12.3 Incremental contribution revisited	14
12.4 Implications for banking and markets	14
12.5 Limitations	14
12.6 Takeaway	15

1 Big picture

- **Question.** Why do banks and other financial arrangements create liquid, transaction-like securities? The paper asks whether liquidity can arise endogenously as a response to asymmetric information in secondary markets.
- **Core friction.** Some agents need to trade before asset payoffs are realized, but they may be less informed than other traders. If they trade claims whose values depend on private information, informed traders can exploit them.
- **Main mechanism.** Liquidity is created by splitting risky asset cash flows into claims with different information sensitivity. A safe debt claim can be valued without knowing the informed agents' private information; residual equity absorbs the information-sensitive part.
- **Bank debt as a liquid security.** Bank deposits are interpreted as debt claims backed by risky asset portfolios. If deposits are riskless, uninformed agents can use them for transactions without trading at a disadvantage against insiders.
- **Alternative liquid securities.** High-grade corporate debt and government debt can play the same role. The paper therefore sees banks as one important liquidity-creation mechanism, but not the only one.
- **Deposit insurance rationale.** If private bank debt cannot be made riskless, government deposit insurance can create the equivalent safe transactions asset through a tax-subsidy arrangement.
- **Policy implication.** A payments system based on government debt or money market mutual funds holding high-grade debt can, in principle, substitute for a system based on insured bank deposits.

Economic takeaway: Liquidity is not simply low transaction cost. In this paper, a security is liquid when uninformed agents can trade it without losing to better-informed agents. Safe debt is liquid because its payoff is insensitive to the private information that insiders possess.

2 Incremental contribution of the paper

- **A different theory of financial intermediation.** Earlier theories often justify banks through delegated monitoring or risk-sharing. This paper justifies banks through the creation of information-insensitive claims.
- **Liquidity as information insensitivity.** The paper gives a precise economic meaning to liquidity: a security is liquid if uninformed agents can use it for transactions without being adversely selected by insiders.
- **Banks create liquidity by security design.** The bank does not merely pool assets. It issues debt and equity against risky assets, making the debt safe and leaving the residual risk and information sensitivity to equity.
- **A bridge to corporate finance.** The same cash-flow-splitting logic applies to corporate debt. Firms can create relatively liquid securities by issuing safe debt and residual equity, paralleling the logic in Myers and Majluf-style adverse selection settings.

- **A rationale for deposit insurance.** Deposit insurance is not only about preventing runs. It can be seen as a government mechanism for making bank debt riskless when private balance sheets are insufficient.
- **A public-debt interpretation.** Government bonds are another way to create riskless transaction assets. The government's tax power allows it to create safe claims when private contracting may fail.
- **A payments-system implication.** The paper anticipates the idea that money market mutual funds backed by safe short-term debt could serve as transaction media, potentially competing with insured deposits.
- **A foundation for later safe-asset literature.** The paper is an early formal statement of the idea that debt is valuable because it is less information-sensitive than equity.

3 Model setup

3.1 Dates, agents, and preferences

- The economy has three dates, $t = 0, 1, 2$, and a single consumption good.
- There are agents with known preferences at $t = 0$. These agents consume at date 2 and become the **informed traders** because they observe the aggregate state at date 1.
- There are also **liquidity traders**. They do not know at date 0 whether they will need early consumption or late consumption.
- At date 1, a fraction w_i of liquidity traders become early consumers and want to consume immediately. The rest are late consumers and can consume at date 2.
- The fraction of early consumers can be low, w_l , or high, w_h , with $w_h > w_l$.

Interpretation: Liquidity traders need a way to convert long-lived claims into current consumption. This trading motive creates the setting in which asymmetric information becomes costly.

3.2 Endowments and technology

- At date 0, agents receive a capital good. Each unit of capital produces date-2 consumption.
- Capital returns are common across units and can be high, R_H , or low, R_L , with $R_H > R_L > 0$.
- Liquidity traders also receive e_1 units of the consumption good at date 1. This good can be consumed or stored to date 2 at a certain gross return of one.
- Informed agents receive date-2 endowments Me_2 in aggregate and capital endowments M in aggregate.

Interpretation: The model has a single aggregate asset-return risk. This simplification makes the role of debt especially clear: safe debt is produced by placing enough residual equity below it.

3.3 Information structure

- At date 1, informed traders observe both the asset return state, R_H or R_L , and the aggregate liquidity state, w_h or w_l .
- Liquidity traders observe only their own preference realization: early or late. They do not directly observe the aggregate state or asset payoff state.
- Prices may reveal some information, but informed traders can coordinate trading strategies to make some states appear similar to uninformed traders.

Interpretation: The information asymmetry is not about borrowers hiding effort or firms hiding quality at issuance. It is about secondary-market trading when some agents must trade and others know more about asset values.

4 Full-information stock-market benchmark

Under full information, all agents know the realized capital return and the realized fraction of early consumers. The key object is the date-1 price p_{ij} of one unit of capital in terms of the consumption good, where $i \in \{h, l\}$ indexes the fraction of early consumers and $j \in \{H, L\}$ indexes the capital return.

Market clearing requires

$$Nw_i p_{ij} \leq N e_1 (1 - w_i).$$

If late consumers sell all of their date-1 consumption good, then there is no storage and

$$p_{ij} = \frac{e_1(1 - w_i)}{w_i}.$$

If late consumers store some of their consumption good, they must be indifferent between storage and holding capital, so

$$p_{ij} = R_j.$$

The paper assumes

$$R_H > \frac{e_1(1 - w_l)}{w_l} > \frac{e_1(1 - w_h)}{w_h} > R_L.$$

Interpretation: With full information, liquidity trading is efficient. Early consumers sell capital to late consumers at a price that reflects true fundamentals. Informed agents have no special trading advantage because everyone knows the state.

Economic message: The benchmark shows what is lost under asymmetric information. The problem is not that trade is impossible; the problem is that prices may not fully reveal the state when insiders strategically supply assets.

5 Asymmetric information and insider trading

5.1 The insider coalition

When liquidity traders are restricted to holding stock, informed agents can exploit them. Early consumers must sell capital regardless of information. Late consumers decide whether to use their

date-1 endowment to buy capital.

The informed agents can form a coalition and choose how much capital to supply. Their goal is to make some bad states look like good states to uninformed traders. In the key manipulation, the coalition sells capital in the low-return, low-liquidity-demand state $\{l, L\}$ so that the price mimics the high-return, high-liquidity-demand state $\{h, H\}$.

5.2 Proposition 1: insider trading equilibrium

The paper shows that if late consumers' posterior expected return is high enough and informed agents have enough capital to mimic the desired state, then an imperfectly competitive rational expectations equilibrium exists in which prices are fully revealing in only two of four states.

The coalition supplies

$$M_{lL} = \frac{N(w_h - w_l)}{1 - w_h}$$

in state $\{l, L\}$ and makes the price equal to the price that would prevail in state $\{h, H\}$.

Interpretation: The informed coalition takes advantage of the fact that late consumers cannot distinguish a genuinely high-return state from a low-return state with extra insider supply. The equilibrium is self-enforcing because if insiders deviate and reveal the state, the trading profit disappears.

Economic message: An information-sensitive security such as equity is a bad transactions medium for uninformed agents. Its value depends on information they do not have, so they can be picked off by insiders.

6 Private liquidity creation

6.1 Bank intermediation and safe debt

The paper then allows liquidity traders to respond by contracting. A financial intermediary pools capital and issues two claims: debt and equity. Let D be the capital contributed by debt holders and E the capital contributed by equity holders. The intermediary's assets are $A = D + E$.

The debt is riskless if the promised debt payment does not exceed the low-state asset payoff:

$$DR_D \leq AR_L = (D + E)R_L.$$

Equivalently,

$$\frac{D}{D + E} \leq \frac{R_L}{R_D}.$$

Interpretation: The bank creates a safe claim by placing enough equity below the debt. Equity absorbs the information-sensitive residual payoff; debt becomes safe and therefore liquid.

6.2 Breaking the insider coalition

If the ratio of informed agents' capital to liquidity traders' capital, M/N , is sufficiently large, then there exists a promised return R_D such that:

- bank debt is riskless,
- liquidity traders prefer bank debt to stock-market capital,
- informed agents prefer bank equity to membership in the insider coalition.

This destroys the insider coalition. Liquidity traders who need early consumption trade safe bank debt, not stock. Late consumers refuse to buy stock from informed traders because they know informed traders would sell stock only in low-return states.

Economic message: The bank separates transaction services from information-sensitive risk bearing. Uninformed agents hold debt for transactions, and informed agents hold equity for risky investment.

6.3 Intermediary equilibrium

Proposition 3 shows that, if M/N is sufficiently large, an equilibrium exists in which all liquidity traders buy riskless intermediary debt and informed agents provide equity capital. In the limit, the allocation approaches the full-information benchmark.

Interpretation: The bank is not valuable because it hides information. It is valuable because its liability design allows uninformed agents to avoid trading information-sensitive assets.

7 Corporate debt and nonbank liquidity creation

The same logic can operate at the firm level. If firms issue both debt and equity, corporate debt can play the same role as bank deposits. A firm with assets $A = D + E$ can issue safe debt satisfying

$$DR_D = AR_L.$$

Mutual funds can then specialize in holding either debt or equity. In particular, money market mutual funds can hold high-grade debt such as commercial paper and issue shares that function as transaction-like claims.

Interpretation: Banks are one technology for creating liquid claims, but not the only technology. Security design at the firm level can also produce relatively information-insensitive debt.

Economic message: The model gives a theoretical reason why Treasury bills, high-grade corporate debt, commercial paper, insured deposits, and money market fund shares can all be thought of as safe-asset or transaction-media candidates.

8 Government intervention

8.1 Deposit insurance

If bank debt is risky, private debt may not fully protect uninformed agents. Suppose the promised debt payment exceeds the low-state payoff:

$$DR_D > AR_L.$$

In the low state, the government taxes late-consuming agents and uses the proceeds to make depositors whole. In the high state, the bank pays a fair insurance premium. The tax-subsidy scheme can replicate the allocation with riskless private deposits.

The fair insurance premium satisfies

$$\phi = R_D - \frac{D + E}{D} R_L.$$

Interpretation: Deposit insurance is valuable because it transforms potentially information-sensitive bank debt into a riskless transactions asset. The government can do this because it can tax broadly and enforce transfers.

8.2 Government debt market

Government debt plays a parallel role. If the private economy cannot create enough riskless corporate debt, the government can issue bonds backed by future taxes. These bonds become additional safe claims for uninformed agents to use as transaction securities.

Economic message: Deposit insurance and government debt are both ways for the state to create information-insensitive claims. The common element is not banking per se, but the production of a risk-free asset.

9 Theoretical identification and design

- **Benchmark comparison.** The model first defines the full-information allocation, then shows how asymmetric information creates trading losses, and finally shows how contracts restore protection.
- **Mechanism isolation.** The only force creating a role for liquid securities is the information asymmetry in trading. The model does not need risk sharing, monitoring, or bank runs to generate a demand for bank debt.
- **Coalition logic.** Insider trading requires a coordinated informed coalition that can make states observationally similar. Liquidity creation works by removing uninformed agents from the information-sensitive stock market.
- **Contracting logic.** Safe debt is useful because its payoff is independent of the private information about asset returns. Equity absorbs the state-contingent residual.
- **Policy logic.** Government intervention becomes useful only when private safe-claim creation is infeasible or insufficient.

Interpretation: This is not an empirical paper, so identification is theoretical rather than econometric. The paper identifies the role of banks by comparing allocations across regimes: full information, asymmetric information with stock only, private security design, and government-backed safe claims.

10 Extracted paper visuals and original equation panels

Note: The paper contains no formal numbered tables or formal figures. To preserve the visual style of the previous notes and directly use the original paper, this section extracts key original equation and proposition panels from the paper. Each subsection title includes the original object name or substantive label.

10.1 Extract 1: Original title and abstract - information-insensitive securities

Read:

- The abstract states the paper's central claim: trading losses from information asymmetries can be mitigated by splitting cash flows of underlying assets.
- Bank deposits, high-grade corporate debt, and government bonds are listed as examples of liquid securities whose values need not depend on insiders' private information.

Economic message: The paper defines liquidity through protection from adverse selection. A security is liquid when uninformed agents can trade it without being exploited.

10.2 Extract 2: Full-information pricing equations (1)-(5) - no-storage and storage prices

Read:

- Equations (1)-(5) give the stock-market benchmark under full information.
- If late consumers supply all endowment, price reflects market clearing; if they store some endowment, price equals the capital return.

Economic message: Full information makes the price reveal the state. The later adverse-selection problem arises because this clean price signal can be distorted by informed supply.

Financial Intermediaries and Liquidity Creation

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ABSTRACT

Trading losses associated with information asymmetries can be mitigated by designing securities which split the cash flows of underlying assets. These securities, which can arise endogenously, have values that do not depend on the information known only to informed agents. Bank debt (deposits) is an example of this type of liquid security which protect relatively uninformed agents, and we provide a rationale for deposit insurance in this content. High-grade corporate debt and government bonds are other examples, implying that a money market mutual fund-based payments system may be an alternative to one based on insured bank deposits.

(a) Extract 1: Original title and abstract -
information-insensitive securities

state of the world $\{i, j\}$. Market clearing equates the demand for the consumption good with supply. Thus,

$$Nw_i p_{ij} \leq N e_i (1 - w_i). \quad (1)$$

There are two separate cases to consider, one where late consumers sell all of their consumption endowment (condition (1) holds with equality) and one where they sell only part, choosing to store some (condition (1) being strict inequality).

When there is no storage in equilibrium, condition (1) becomes an equality. Solving for the price of the capital good, we have

$$(No\ Storage) \quad p_{ij} = \frac{e_i(1 - w_i)}{w_i}. \quad (2)$$

This case holds under the parametric restriction:

$$R_j > \frac{e_i(1 - w_i)}{w_i}. \quad (3)$$

When storage occurs in equilibrium, late consumers must be just indifferent between buying and holding the capital good and storing the consumption good, i.e.,

$$(Some\ Storage) \quad p_{ij} = R_j. \quad (4)$$

This case holds when the inequality sign in condition (3) is reversed.

Hereafter, we will make the assumption that condition (3) holds for $j = H$, so that, in equilibrium, no storage will occur for the states $\{h, H\}$ and $\{l, H\}$, where the return on capital is high. In addition, we will assume that condition (3) does not hold for $j = L$, so that, in equilibrium, some storage will occur for the states $\{h, L\}$ and $\{l, L\}$, where the return on capital is low. These assumptions can be summarized by the following condition:

$$R_H > \frac{e_i(1 - w_i)}{w_i} > \frac{e_i(1 - w_h)}{w_h} > R_L. \quad (5)$$

(b) Extract 2: Full-information pricing equations
(1)-(5) - no-storage and storage prices

10.3 Extract 3: Proposition 1 - Insider Trading

Read:

- Proposition 1 gives conditions under which an insider-trading equilibrium exists.
- Some prices are fully revealing, but the states $\{h, H\}$ and $\{l, L\}$ can be pooled from the viewpoint of uninformed liquidity traders.

Economic message: When uninformed agents use information-sensitive claims as transaction media, insiders can create pooling prices and extract gains.

10.4 Extract 4: Bank intermediation contract - riskless debt condition (14)-(16)

Read:

- The bank issues debt and equity claims against a pooled asset portfolio.
- Debt is riskless when the promised debt payment is no larger than the low-state asset payoff.

Economic message: The safe part of the capital structure is liquid because it can be valued without private information. Equity is the residual information-sensitive claim.

Let $R^* = q_h' R_h + q_l' R_l$ be the uninformed late consumers' expectation at time 1 of the return on capital when state $\{l, L\}$ actually occurs, where q_h' and q_l' are their posterior probabilities of the states being $w_l = w_h$ and $w_l = w_l$ respectively. The following proposition demonstrates the existence of insider trading by the informed agents.

PROPOSITION 1 (Insider Trading): Let $\{\hat{p}_{ij}\}$ be the full-information prices for states, $\{i, j\}$. If (i) $e_1(1 - w_h)/w_h \leq R^*$ and (ii) $\frac{M}{N} \geq \frac{(w_h - w_l)}{(1 - w_h)}$, then there exists Imperfectly Competitive Rational Expectations Equilibrium prices $\{p_{ij}^*\}$, where $p_{lH}^* = \hat{p}_{lH}$, $p_{hL}^* = \hat{p}_{hL}$ and $p_{hH}^* = \hat{p}_{hH}$. That is, these prices are fully revealing in only two of the four states.

Proof: We will verify that the following specification of prices and strategies constitutes an equilibrium for the assumed parameter values.

State $\{l, H\}$

$$p_{lH}^* = \frac{e_1(1 - w_l)}{w_l}, M_{lH} = 0; S(p_{lH}^*) = N(1 - w_l)e_1 \quad (\text{No Storage}).$$

State $\{h, L\}$

$$p_{hL}^* = R_L; M_{hL} = 0; S(p_{hL}^*) < N(1 - w_l)e_1 \quad (\text{Some Storage}).$$

State $\{h, H\}$

$$p_{hH}^* = \frac{e_1(1 - w_h)}{w_h}, M_{hH} = 0; S(p_{hH}^*) = N(1 - w_l)e_1 \quad (\text{No Storage}).$$

(a) Extract 3: Proposition 1 - Insider Trading

A. Bank Intermediation and Liquidity Creation

Suppose at date $t = 0$ the following contract is offered to agents. An intermediary will be set up which pools agents' capital and issues securities to them. Let $A = N_I + M_I$ be the total endowment of the capital good contributed by members of this intermediary as of date $t = 0$, where $N_I = N - N_S$ and $M_I = M - M_S$. The subscript I refers to the capital of agents joining the intermediary, and S refers to the capital of agents continuing to invest in the stock market. The total return of the intermediary's assets at date $t = 2$ is AR_i , $i = H, L$. Ownership of two types of claims on this capital is offered to agents: debt claims and equity claims. Let D and E (whose sum equals A) be the total amount of capital contributed by agents who own debt and equity claims, respectively.

The contract also imposes a debt-to-equity ratio ceiling such that the total payment promised to debt claim, DR_D , must be less than or equal to AR_L ; i.e., debt claims are required to be riskless:

$$DR_D \leq AR_L = (D + E)R_L. \quad (14)$$

Therefore,

$$\frac{D}{(D + E)} \leq \frac{R_L}{R_D} \text{ or } E \geq \frac{D(R_D - R_L)}{R_L}. \quad (15)$$

We would like to consider whether offering agents this intermediary contract would affect the Imperfectly Competitive Rational Expectations Equilibrium analyzed in the previous section. Before stating a series of propositions related to this issue, we make an additional assumption that will simplify the proof of the first of these propositions. We assume that conditional on being a late consumer, the probabilities of the state being $w_i = w_h$ or $w_i = w_l$ are equally likely. If late consumers form expectations in a Bayesian manner, this implies

(b) Extract 4: Bank intermediation contract - riskless debt condition (14)-(16)

10.5 Extract 5: Proposition 2 - Nonexistence of stock-market insider equilibrium

Read:

- Proposition 2 shows that a small group of liquidity traders can form a bank if informed capital is large enough relative to liquidity-trader capital.
- Informed agents may prefer bank equity to staying in the insider coalition.

Economic message: The bank contract breaks the coalition by giving insiders an alternative residual claim and giving liquidity traders a safe transaction claim.

10.6 Extract 6: Proposition 3 - Existence of an intermediary equilibrium

Read:

- Proposition 3 establishes an equilibrium in which all liquidity traders hold riskless intermediary debt and informed agents contribute equity.
- The stock market becomes a lemons market for liquidity traders, who therefore avoid it.

Economic message: Intermediation restores protection for uninformed agents by changing what they trade, not by making them better informed.

$$q_h(1 - w_h) = q_l(1 - w_l). \quad (16)$$

Now suppose that liquidity traders are allowed to offer the intermediary contract to all agents as a possible trading mechanism. It is clear that, for R_D sufficiently high, liquidity traders are better off holding bank debt. The question is whether the informed agents can be induced to defect from the Insider Coalition to become the bank's equity holders. If this occurs, the intermediary contract will be feasible and the equilibrium of the previous section will not exist.

PROPOSITION 2 (Nonexistence of Stock Market Insider Equilibrium): *Consider a small number of liquidity traders, say N_l (close to zero), choosing to form a bank. Then, if the ratio of informed to uninformed agents' capital, $\frac{M}{N}$, is sufficiently large, there exists a rate of return on intermediary debt, R_D , such that (i) debt is riskless, (ii) liquidity traders prefer to invest their capital in the debt of the intermediary rather than the stock market, and (iii) individual informed agents prefer to invest their capital in the equity of the intermediary rather than the stock market insider coalition.*

(a) Extract 5: Proposition 2 - Nonexistence of stock-market insider equilibrium

...the smaller is the required rate of return on bank equity necessary to induce an individual insider to join the bank and defect from the Insider Coalition. Consequently, if the required return on bank equity is not too large, the rate of return on bank debt will be large enough to attract an individual liquidity trader away from the stock market as well.

The next proposition states that an equilibrium can exist where *all* liquidity traders choose to purchase the riskless debt of an intermediary and informed agents derive no advantage from operating an Insider Coalition in the stock market. The proof of this proposition assumes the following condition, which includes condition (5) assumed previously:

$$R_L < \frac{e_l(1 - w_h)}{w_h} < \bar{R} < \frac{e_l(1 - w_l)}{w_l} < R_H. \quad (17)$$

PROPOSITION 3 (Existence of an Intermediary Equilibrium): *If $\frac{M}{N}$ is sufficiently large, then there exists an equilibrium where (i) all liquidity traders purchase riskless debt of the intermediary and (ii) informed agents will choose to contribute equity capital.*

Proof: See the Appendix.

The intuition behind this result is that, if informed agents' capital is sufficiently large relative to that of the liquidity traders, it is feasible for a bank to issue sufficient riskless debt that can be used by all liquidity traders for transactions.³

(b) Extract 6: Proposition 3 - Existence of an intermediary equilibrium

10.7 Extract 7: Corporate debt and government-debt introduction - nonbank liquidity creation

Read:

- The paper explicitly notes that corporate debt can replicate bank-created safe debt when firms split risky cash flows into debt and equity.
- The same page introduces deposit insurance and government debt markets as ways of creating risk-free assets.

Economic message: Banks are not unique. Corporate debt, debt mutual funds, and government debt are alternative institutional arrangements for creating liquid safe claims.

10.8 Extract 8: Proposition 4 - Deposit Insurance

Read:

- Proposition 4 shows that when bank debt is risky, a government tax-subsidy scheme can implement the same expected utility as riskless private bank deposits.
- The government can create risk-free claims through its taxing capacity.

Economic message: Deposit insurance is valuable because it makes deposits information-insensitive. The role of the government is to complete safe-asset creation when private balance sheets cannot.

B. Corporate Debt and Liquidity Creation

So far we have implicitly assumed that “firms” do not issue debt. That is, when we considered the stock market equilibrium in Section III, we imagined individuals exchanging their capital with firms who issued them equity shares. In this section we briefly consider what happens if the firms are willing to buy capital at $t = 0$ in exchange for either debt or equity. So now there exists a market for corporate debt, such as commercial paper.

Suppose a firm offers to pay R_D per dollar of debt and issues an amount of riskless debt such that $DR_D = AR_L$, where $A = D + E$ is the firm's total assets. Then it is immediately apparent that the firm can offer the same riskless debt as the bank intermediary we described previously. All of the above arguments about the bank now apply to the firm. Agents need not directly hold the claims of firms, but mutual funds could arise to specialize in holding either debt or equity claims. In particular, funds similar to money market mutual funds could purchase the high-grade debt (e.g., commercial paper) of firms. As before, the equilibrium would be for all liquidity traders to buy claims on the debt fund and all informed traders to buy claims on the firm's equity. We comment further on this in our concluding remarks.

V. Deposit Insurance and a Government Debt Market

A deposit insurance system for banks can also satisfy the liquidity traders' desire for a safe asset for trading. In this section we show how deposit insurance can replicate the allocation of the previous section when intermediary debt is risky. In addition, we show that development of a government debt market is similar to deposit insurance, as it involves government creation of a risk-free asset. In a like manner, a government debt market can replicate the riskless corporate debt contract when riskless corporate debt is in insufficient supply. The government can succeed where private contracting fails due to its ability to enforce lump sum taxation. It is the revenue from this taxation that accounts for the government's

(a) Extract 7: Corporate debt and government-debt

introduction - nonbank liquidity creation

10.9 Extract 9: Conclusion - banks, safe debt, Treasury securities, and money market funds

Read:

- The conclusion summarizes the main claim: by splitting risky cash flows, liquid securities eliminate the advantage of better-informed agents.
- It also discusses Treasury securities and money market mutual funds as potentially important transaction media.

Economic message: The broad lesson is a safe-asset lesson: transaction media must be robust to asymmetric information. Banks, governments, and debt mutual funds are different ways of producing such claims.

64

The Journal of Finance

which implies that

$$\phi = R_D - \frac{(D + E)}{D} R_L. \quad (21)$$

Substituting (21), the expression for the fair deposit insurance premium, into (19) yields

$$E[R_E]E = R(D + E) - R_D D. \quad (22)$$

As in the previous section, consider a competitive equilibrium where the expected rate of return on equity approaches $\bar{R}$. In this case, equation (22) shows that R_D also approaches $\bar{R}$. Therefore, the allocation under the deposit insurance scheme gives agents the same expected utility as in the case of the private uninsured intermediary considered in the previous section. In summary, we have shown the following.

PROPOSITION 4 (Deposit Insurance): *When bank debt is risky, the tax-subsidy scheme $\{T, \phi\}$, defined above, can implement an allocation which gives all agents the same expected utility as in the riskless private bank deposit allocation.*

Similar to government intervention as a deposit insurer, we can consider whether government intervention can benefit uninformed agents when firms issue corporate debt, as was described previously. Let us start from the assumption that each firm issues riskless debt such that

$$A_i R_L \geq D_i R_{D_i}, \quad (23)$$

where A_i and D_i are the assets and debt of firm i , respectively. However, suppose

(b) Extract 8: Proposition 4 - Deposit Insurance

VI. Conclusion

The historically popular notion that informed agents can benefit at the expense of uninformed agents is true in the setting which we have analyzed. Informed agents can form an insider coalition which is self-enforcing and can benefit at the expense of the lesser informed agents. When this condition exists, a demand for liquid securities by uninformed agents will result. By splitting risky cash flows, liquid securities are created which have the effect of eliminating the potential advantage possessed by better informed agents.

Liquidity can be created through the formation of banks. We have formalized a traditional rationale for the existence of banks and deposit insurance, namely that they provide a riskless transactions medium that eliminates the need of uninformed agents to trade in assets whose returns are known by better informed agents. By issuing deposits, banks create “riskless” securities for trading purposes. In instances where bank asset risk is such that uninsured deposits cannot be made riskless, we have shown that deposit insurance can replicate the allocation achieved with riskless private bank deposits.⁶ In addition, liquid securities can also be created through the formation of corporate debt or government securities markets. As an alternative to bank intermediation, firms can split risky cash flows, thereby creating a safer security (debt).

An empirical implication of our model is that transactions securities should be the most actively traded assets. This is consistent with the relatively high turnover in ownership of insured bank liabilities and Treasury securities. Corporate debt, on the other hand, is much less actively traded, suggesting that our assumption that firms can create riskless securities simply by splitting the cash flows of their underlying assets is not completely accurate.

For tractability, we studied a model with a single source of asset risk. Clearly, with multiple sources of asset risk, diversification would provide another, perhaps complementary, channel for the reduction of risk. This channel implies combining imperfectly correlated assets to reduce risk, rather than splitting cash flows. These issues are investigated in Gorton and Pennacchi (1989). The creation of mutual funds holding a diversified portfolio of corporate debt can alleviate the inability of individual firms to create riskless debt. For example, money market mutual funds are large holders of commercial paper, and the shares of these funds provide a potentially important transactions medium.⁷

⁶ An issue which we have not considered concerns possible equilibria where banks exist but their

Extract 9: Conclusion - banks, safe debt,

Treasury securities, and money market funds

11 Short summary

- In a stock market with full information, liquidity traders can trade capital efficiently.
- With asymmetric information, informed traders can form a coalition and manipulate the supply of capital to make low-return states look like high-return states.
- Uninformed liquidity traders lose because equity is information-sensitive.
- Banks solve the problem by issuing riskless debt and residual equity against risky assets.
- Bank deposits are liquid because their payoff does not depend on the private information held by insiders.
- Corporate debt, Treasury securities, and money market mutual fund shares can also serve as information-insensitive transaction assets.
- Deposit insurance is justified when private bank debt cannot be made riskless; it transforms risky deposits into safe claims through government backing.

12 Conclusion

12.1 Core conclusion

Gorton and Pennacchi argue that financial intermediaries create liquidity by designing claims that protect uninformed agents from adverse selection. The key object is not the bank's asset portfolio alone, but the liability structure: safe debt plus risky residual equity.

12.2 Economic intuition

The problem is that some agents need to trade, but they do not know asset values as well as insiders. If they trade equity, they can be exploited. If they trade a safe debt claim, they do not need to know the state because the claim pays the same amount across states. Liquidity therefore comes from information insensitivity.

12.3 Incremental contribution revisited

The paper contributes a theory of banks, deposits, corporate debt, deposit insurance, and government debt under one unifying principle: safe claims are valuable because they are robust to information asymmetry in trade. This distinguishes the paper from bank theories based only on delegated monitoring or Diamond-Dybvig-style risk sharing.

12.4 Implications for banking and markets

- Bank deposits are liquid because they are debt claims made sufficiently safe.
- Deposit insurance can be justified as a way to protect transaction users when private bank debt is not riskless.
- High-grade corporate debt and Treasury securities can also be transaction securities.
- A money market mutual fund-based payments system is a theoretical alternative to a bank-deposit-based payments system.

12.5 Limitations

- The model uses a single source of asset risk, so diversification is not the central mechanism. With multiple risks, diversification and cash-flow splitting would interact.
- The insider coalition is assumed to be feasible and self-enforcing under specified conditions; real-world insider coordination may be more complex.
- The government intervention analysis abstracts from moral hazard, political economy, and distortionary taxation.
- The paper is theoretical and does not provide direct empirical tests of turnover, spreads, or payments-system use.

- The safety of private debt is more difficult in practice than in the model, which the authors acknowledge when discussing less active trading in corporate debt.

12.6 Takeaway

The paper's core message is that liquidity is created when financial claims are made safe enough to be traded without information. Banks, corporate issuers, money market funds, and governments all create liquidity by producing claims whose values are not very sensitive to private information. In this sense, the paper is an early foundation for the modern view of safe debt as information-insensitive money-like assets.